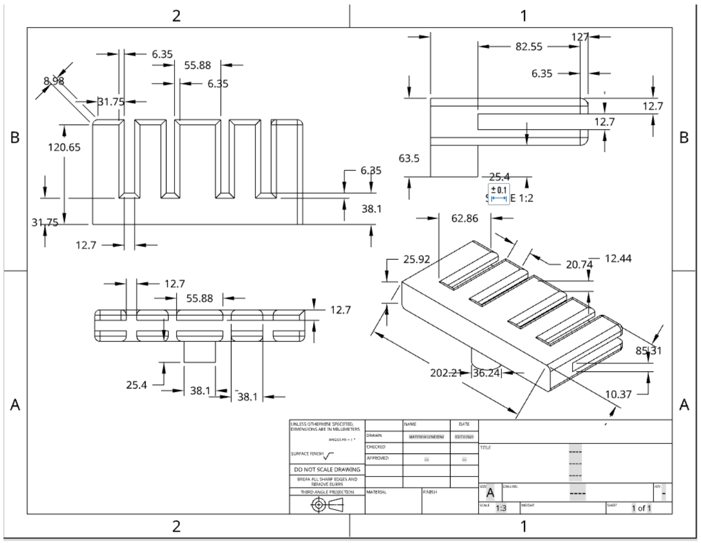
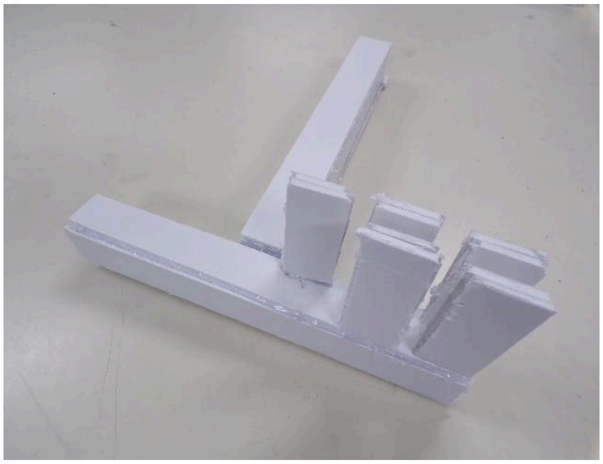
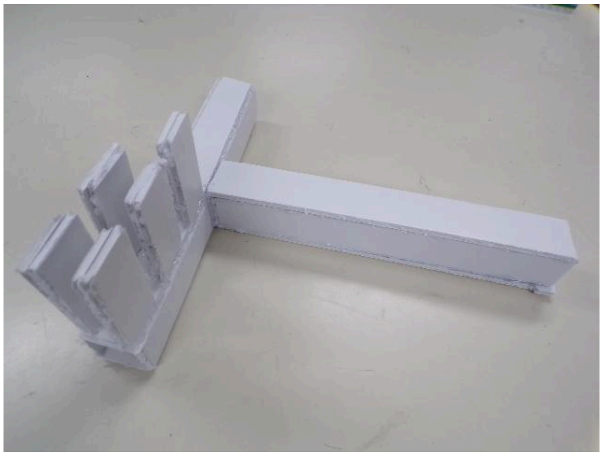
Element G: Construction of a Testable Prototype

Prototype Construction of Design #1 **Foam-Cord - Initial Prototype**



Sketch and dimensions of Six Finger Clamp (OnShape CAD Drawing)





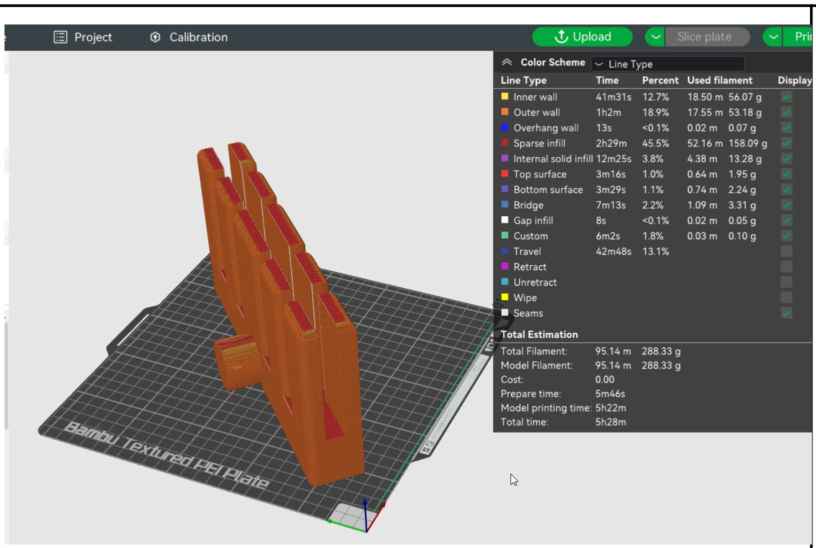
Top Angle (Same concept as 1st Photo)

Partially built to get the idea of size	FInished design of prototype #1 (Only had done to move on to onshape design and main idea)

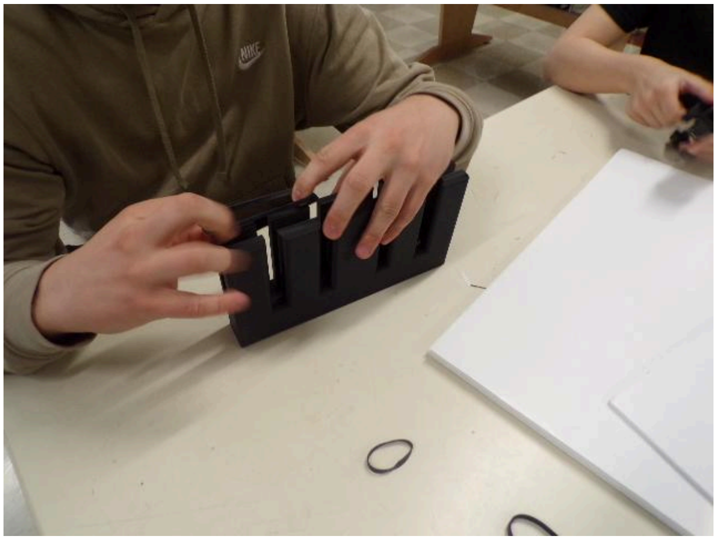
Details about all steps of construction:

	Details	Tools and Manufacturing Methods used
Step 1	Create design through onshape	Computer
Step 2	Use a ruler to measure dimensions that align with the onshape sketch. Also used a pencil on foam cord to be neat while getting dimensions	Ruler, Pencil
Step 3	Cut out foam cord molds for the model	Foam Cord, boxcutter
Step 4	Glue tubes together perpendicularly	Hot Glue gun
Step 5	Attach “fingers”	Foam cord, box cutter, hot glue

Prototype Construction of Design #2 PLA Filament



1. OnShape Design on Bambu app being 3D Printed (Image shows information on how long it took, materials used, and more.)



2. After 3D Printing, put rubber bands around the fingers used as holders so the reflector board wouldn't move.

3. After finishing putting the rubber band around the Finger Holder it was ready to clamp on. This model is 194 grams.

4. Clamp and Holder connected together(fit perfectly)

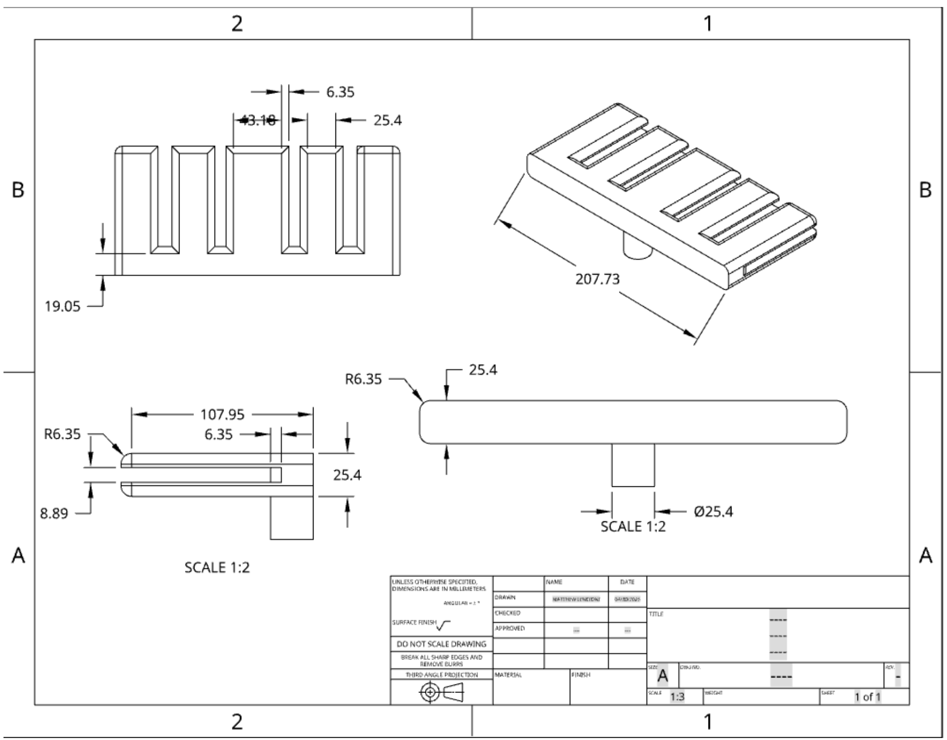
<table>
<tr>
<td>
7. Same concept as #6 just different angle</td>
<td>
8. Same concept as #6 just different angle</td>
</tr>
<tr>
<td colspan="2">
9. Weight the new model to see how it compared to the 2nd prototype. Ended up being way more light with it being 96.3 grams compared to the 2nd Prototype which was 194 grams.</td>
</tr>
</table>

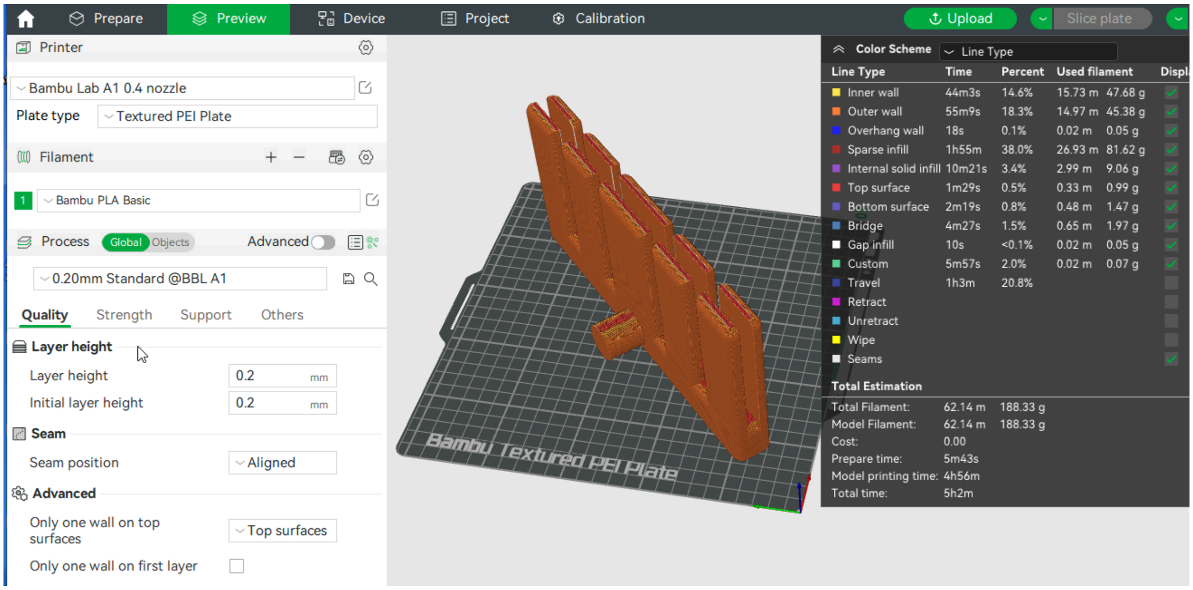
Details about all steps of construction:

	Details	Tools and Manufacturing Methods used
Step 1	Create 3d model in onshape	computer
Step 2	Transfer file to a 3d printer for printing	3d Printer
Step 3	Put rubber bands on model for grippiness	Rubber Bands

Prototype Construction of Design #3 PLA Filament

Sketch and dimensions of Six Finger Clamp Configure #3 (OnShape CAD Drawing)

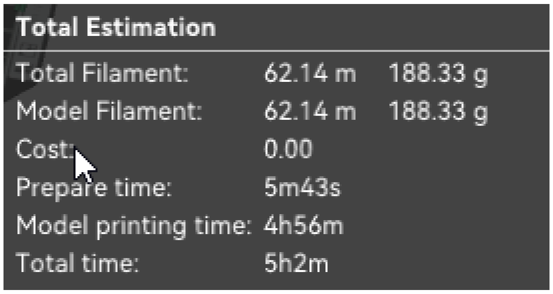




1. Clamp Design #3 on Bambu slicing app. Configuration to 3D Printed (5 hours, 62.14m PLA Filament)

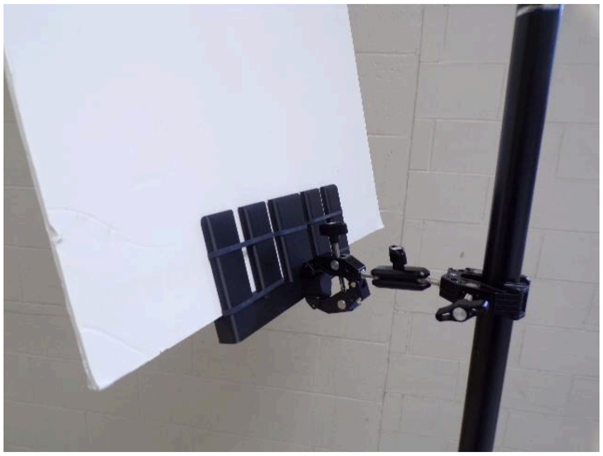
1. OnShape Design on Bambu app being 3D Printed (Image shows information on how long it took, materials used, and more.)

(See below)

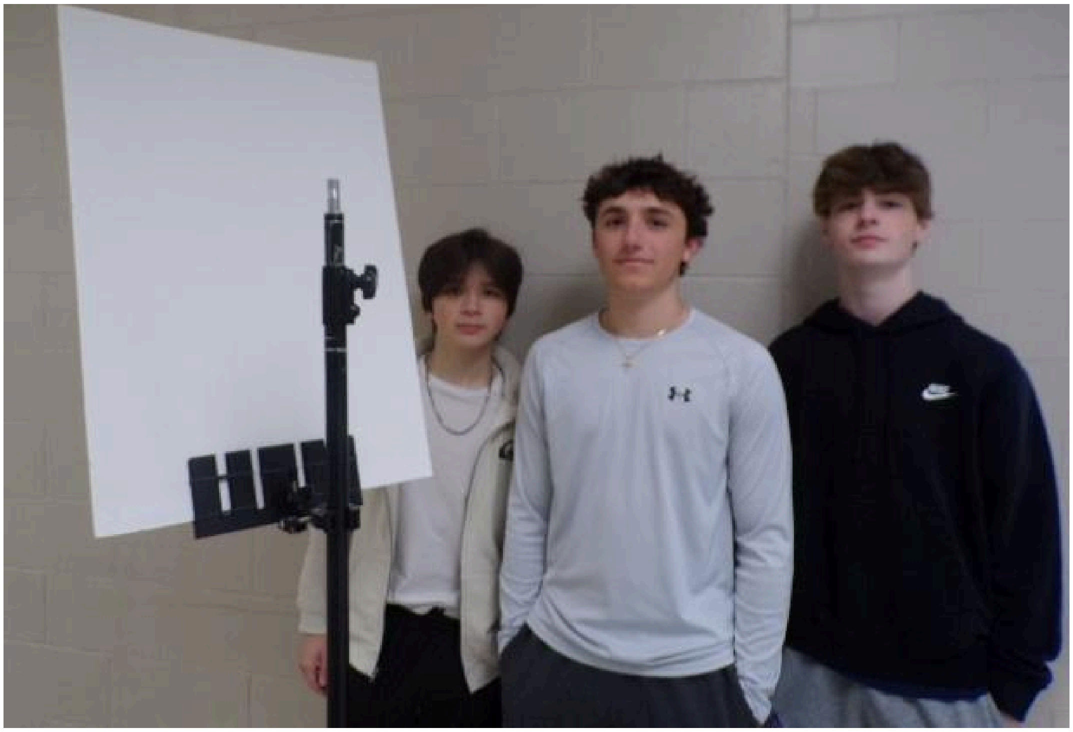


Total Estimation		
Total Filament:	62.14 m	188.33 g
Model Filament:	62.14 m	188.33 g
Cost:	0.00	
Prepare time:	5m43s	
Model printing time:	4h56m	
Total time:	5h2m	

2. measured the weight of the third prototype and it was a success because we tried to make it lighter than the last

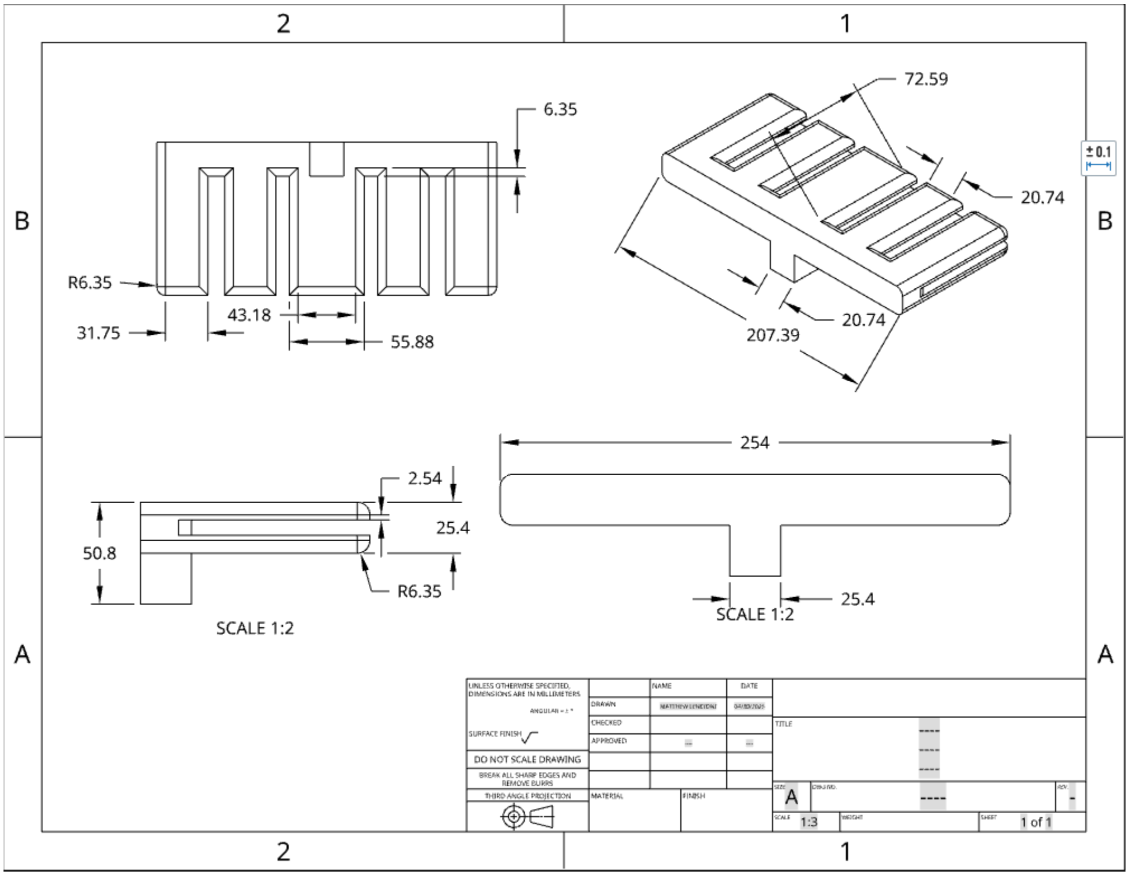
	prototype
 3. Tested the prototype to see if there are any design errors and make sure it fits our requirements.	 4. We then got reviewed by multiple professional engineers and took notes for our design.

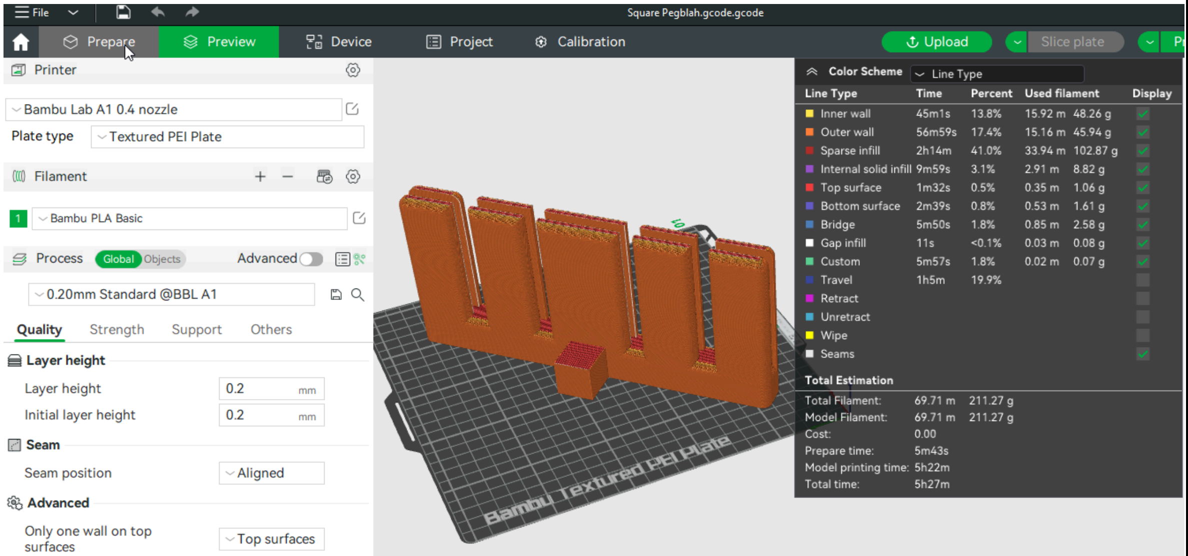
	Details	Tools and Manufacturing Methods used
Step 1	Shrink model and adjust it in onshape	computer
Step 2	Transfer file to a 3d printer for printing	3d Printer
Step 3	Put rubber bands on model for grippiness	Rubber Bands



Prototype Construction of Design #4 PLA Filament - Best Prototype

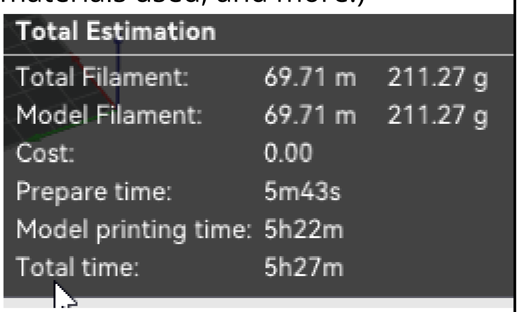
Sketch and dimensions of Six Finger Clamp Configure #4 (OnShape CAD Drawing)





1. Clamp Design #4 on Bambu slicing app. Configuration to 3D Printed (5 hours 27 min, 69.71m PLA Filament)

1. OnShape Design on Bambu app being 3D Printed (Image shows information on how long it took, materials used, and more.)



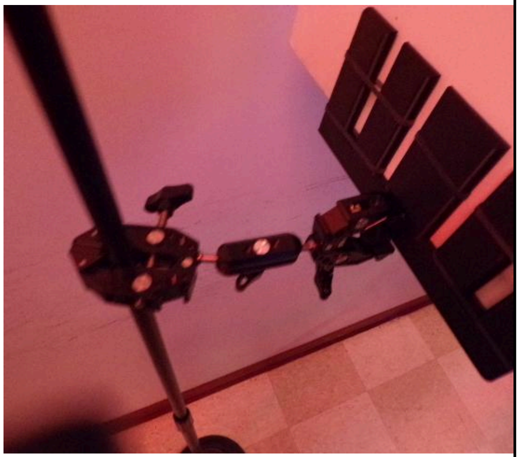
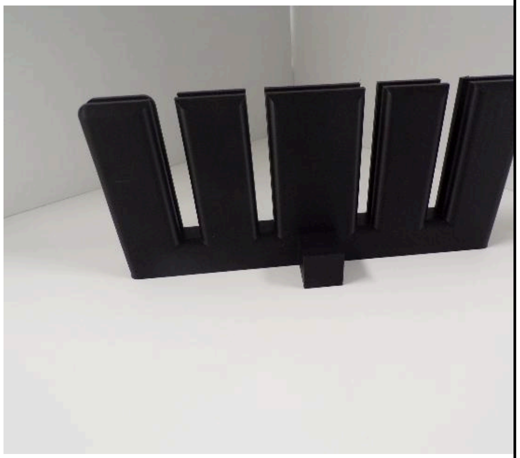
Total Estimation

Total Filament:	69.71 m	211.27 g
Model Filament:	69.71 m	211.27 g
Cost:	0.00	
Prepare time:	5m43s	
Model printing time:	5h22m	
Total time:	5h27m	

2. We measured the design and it turned out to weigh more than prototype 3 but less than 2 so it was a success.

3. We then tested the new square nub that we added onto our design if it passed all of the requirements and

4. We then reviewed with Mr. O'farrell to see if he approved our design and he said it was perfect.

worked well with the clamp. 	

	Details	Tools and Manufacturing Methods used
Step 1	Adjust “nub” in onshape from previous models file	computer
Step 2	Transfer file to a 3d printer for printing	3d Printer

Material name	**Description**	**Where material was purchased**	**Cost per unit**	**# units**	**Total cost per item**
PLA Filament	Material used for 3D printing	Through the school	$4.00	1	$4.00

3D Printer	Printer that uses PLA filament to create 3d objects	Through the School	$599.00	1	$599.00
Dual Clamp	Mechanism allowing for two items to be clamped onto	Amazon	$19.99	1	$19.99
Rubber Bands	Material used to make reflector board stay steady while on clamp	Through the school	$1.90 for 500 bands	1	$1.90

Construction of the Final Prototype *(G2)*

Design Requirement	How will this prototype allow you to collect objective, measurable data?
Adjustability	Use the clamp to switch the angle at which it is at to determine is it adjusts smoothly
Clamp must be secure	Ensure that the clamp fastens tight enough
Must be stable	Ensure that once fastened the clamp minimizes movement of the finger holding mechanism
Must be portable	Ensure that the clamp and mechanism can fit into a certain size bag or something else to transport the clamp

Testing or Modeling of Attributes *(G3, G4)* **and Addressing Non-Testable Attributes** *(G4)*

Attribute	**Description**	**Describe your prototype's**

		functionality to test this attribute
Finger Molds	The finger molds are used to support the board and hold it in place	Expert Review Required
Rubber Bands	The rubber bands are utilized to create friction, which helps keep the board in place	Expert Review Required
"Stub"	The stub is important because it allows for the clamp to easily attach to the finger holding mechanism	Can be tested to see if it is able to be clamped onto and gripped.
Dual Clamp	The clamp is vital as it connects the finger holding mechanism to the pole in which is needed	Can be tested by seeing if the clamp attaches to the "stub".

Addressing Non-Testable Attributes *(G4)*

For all attributes that you listed "expert review required" in the table above, provide a justification for why that is true. "Field experts" are professionals who spend at least a portion of their time working in a field directly related to your project. These field experts are not permitted to double count as stakeholders.

Attribute	**Justification for why this attribute cannot be modeled or tested with your prototype**
Finger Molds	We need an expert to see if this is the most efficient model to hold the board
Rubber Bands	We need an expert to see if this grip mechanism can be modified or changed.